

Stakeholder

18+3 · WEIGHT 5 · LONGEST

Open Franka in-the-wild flywheel — 76k / 350h / 564 scenes — buy ~+20% robustness without 22 embodiments.

Paper club · Data cluster · due 9:30 AM PT · Midterm Nov 6 nearby

Spine: Same Franka · many scenes · protocol > vibes · Fig.10 before Fig.8 · 2-16-8 reactive · steal protocol into BEHAVIOR

Khazatsky et al. · 13 labs · 18 identical Franka replicas · CC-BY 4.0 · droid-dataset.github.io

Talk arc — Stakeholder densest

Product → moats → decisions → motion → risks → exam

01-03 Flywheel + OXE contrast

76k/350h/564 vs many-robots

04-06 Who pays / moats

564/52 · +22/+17 · π_0 9.1%

07-09 Nov 4 → midterm tree

Franka week / VLA kill-switch / BEHAVIOR

10-13 Motion + long-horizon

2-16-8 · Cook Lentils · Fig.10→Fig.8

14-16 Risks + exam bite

lock-in · OpenVLA drop · tattoo

17-18 Nov 6 card + close

numbers + one-liner

Product flywheel — same Franka, many scenes

~~Distributed Robot Interaction Dataset~~

76k

successful trajectories

350 h

teleop hours

564

authentic scenes

52

buildings

86

verbs / tasks

50

collectors @ 13 labs

One public Franka+Robotiq+ZED stack × 18 replicas · Quest teleop · Polymetis 15 Hz · CC-BY 4.0
~16k fails released but excluded from 76k headline and paper co-train mixes

OXE contrast — many robots vs many scenes

Opposite axes of diversity

OXE

Many robots / aggregated lab pools

- 22 embodiments
- ~311 scenes (agg, many lab-like)
- Municipal water — breadth
- Great for who the robot is

DROID

Same Franka · many real scenes

- 1 public research stack
- 564 scenes / 52 buildings
- Protocolized scene churn
- Great for where it works

Moral: OXE breadth $\neq$ substitute for authentic scene entropy on the eval embodiment.

Stakeholder buy: co-train DROID with a handful of site demos — don't buy 22 embodiments first.

Who pays / who benefits

Buyers, free-riders, non-consumers

Franka labs

- Clone cart (hardware guide)
- Add 50-150 local demos
- Co-train 50/50 before new arch

Foundation teams

- Scene-diversity shard in mix
- Mix-bug canary (OpenVLA drop)
- π_0 9.1% / pi05_droid path

BEHAVIOR org / entrants

- Scene count $\times$ protocol moves OOD
- Even when traj $\ll$ OXE/RT-1
- Steal churn, don't re-drink river

Non-consumers

- No Franka/ZED budget
- Need bimanual / mobile-base
- Desk/room-centric single-arm

Moats — exact numbers

What you can cite on Nov 6

Scene moat 564 scenes / 52 buildings vs BridgeV2 24

Calibration Extrinsics + App. G $\sim 36k$ cam $\rightarrow$ base / curated $\sim 24k$

Co-train lift +22% ID / +17% OOD absolute (Fig. 8, vs next-best)

Causal diversity Fig. 10: 7362 diverse > 7362 top-20 scenes (OOD 0.60 vs 0.40)

Ecosystem Inside OXE · π_0 open mix 9.1% includes DROID · OpenPI pi05_droid · FAST ZS

Protocol GUI random tasks · $\leq \sim 20$ min / ~ 100 traj scene churn · view/lighting augs

Decision tree — Nov 4 → midterm Nov 6

Four forks; pick with numbers

1 Have Franka + a week of teleop?

Collect 50-150 demos → paper DP or modern chunk head BEFORE new architecture. Cite +22/+17 and 564 scenes.

2 Mixing into a VLA?

Treat DROID as a monitored shard — watch action loss; pre-define kill switch (OpenVLA loyalty lesson).

3 Designing BEHAVIOR data?

Copy protocol: random prompts, scene time-caps, view/lighting jitter, success flags — not vibes / another download.

4 Budget truth?

13 institutions × 12 months × 50 collectors is the hidden cost; open release socializes it. One building ≠ Fig. 8.

Protocol > vibes — steal these knobs

Methodology is the main scientific product

1 Move the robot

≤~100 traj / ~20 min per scene → forced turnover

2 GUI random tasks

From collector-authored scene list → cuts easy-task bias

3 Scene augmentations

Nudge base, remount cams, lighting, add/remove clutter

4 Multi-instruction labels

≤3 / episode crowdsourced post-label (tasq.ai)

5 Success flagging

At collect time; ~16k fails retained, excluded from co-train

6 Shared portable stack

Public hardware guide → community multiplier ×18

Motion deep dive — chunked reactive BC

NOT a classical planner / MPC / skill graph

FLAG: Diffusion Policy · obs=2 / pred=16 / action=8 @ 15 Hz

commit_s = 8/15 $\approx$ 0.53 s open-loop then replan · Absolute EE + gripper · ResNet-50 + frozen DistilBERT (pre-VLA)

Obs horizon	2	Short visual/proprio history
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Prediction	16 @ 15 Hz	$\approx$ 1.07 s planned chunk
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Exec / action	8 open-loop	Receding-horizon chunking
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Cams in policy	2 exterior @ 128²	Wrist collected but underused
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Family	DDPM/DDIM DP	Robomimic backbone; denoising imitation
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Long-horizon stress — Cook Lentils / Clean Desk

Stage completion under OOD, not hierarchical planning

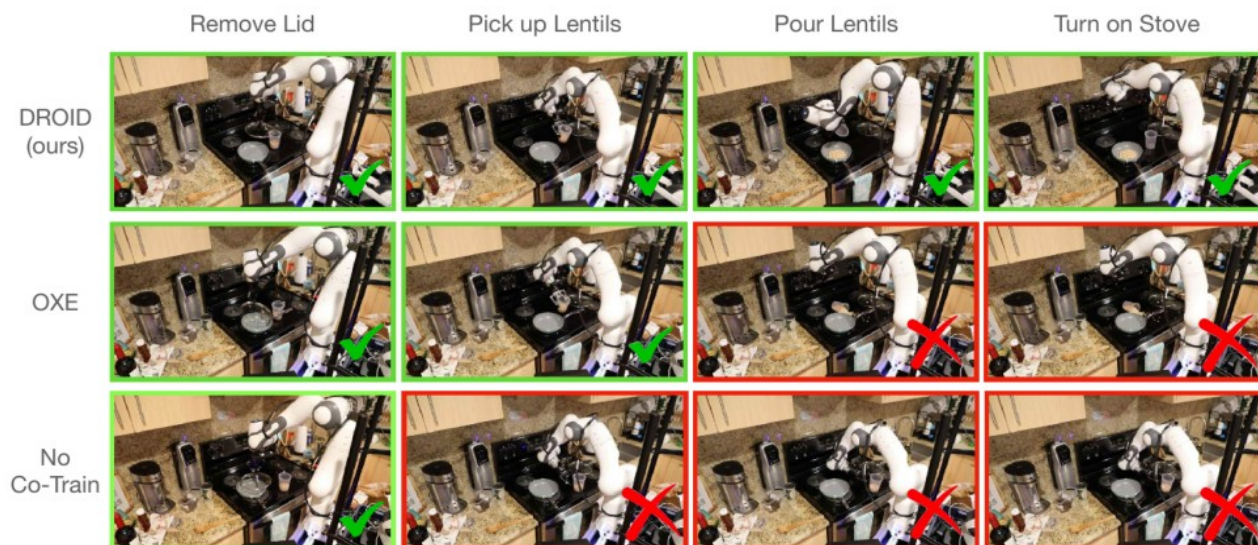


Fig. 9: Representative policy rollout examples on the most challenging “Cook Lentils” task in the kitchen of one of the authors. Qualitatively, we find that policies co-trained with DROID perform smoother, more precise motions, allowing them to solve long-horizon tasks like lentil cooking even in the presence of unseen distractor objects. In contrast, policies trained only with in-domain demonstrations or co-trained with Open-X data [39] struggle with long-horizon tasks and out-of-distribution evaluation settings. See <https://droid-dataset.github.io> for rollout videos.

Why it matters

- Cook Lentils: pour → pan → stove (all 3 stages)
- Clean Desk: stage-rich office cleanup
- DROID co-train finishes stages more reliably under OOD
- No Co-train / OXE struggle after 1-2 stages (Fig. 9)
- Gains = diverse co-train + chunked BC
- NOT a hierarchical planner win
- 50 in-domain demos each + 50/50 mix

Fig. 10 FIRST — diversity drives OOD

Matched 7362 traj: diverse > top-20 scenes

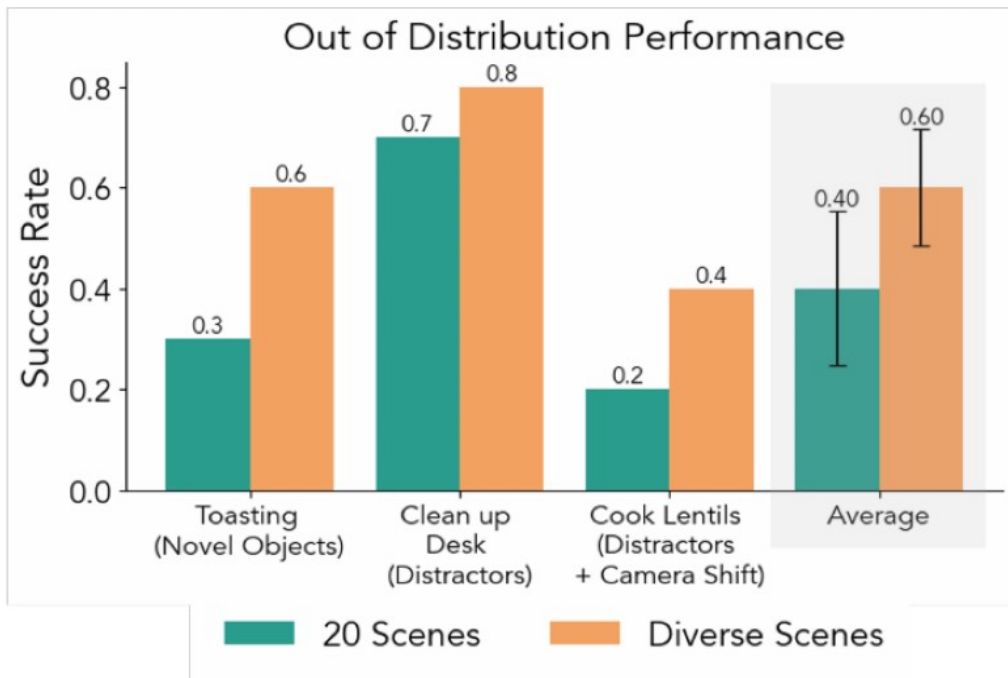


Fig. 10: **How important is the scene diversity in DROID?** We find that co-training on a subset of DROID with diverse scenes has higher OOD performance than co-training on a subset of DROID with only 20 scenes, suggesting the scene diversity of DROID is one of the

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Causal-ish claim

- Fixed N = 7362 trajectories
- Top-20 scenes vs diverse sample
- OOD avg ~0.40 vs ~0.60
- Diversity — not just hours — drives OOD
- Cite THIS before aggregate Fig. 8
- Composition of scenes beats raw demo count

Fig. 8 — aggregate co-train lift

+22% ID / +17% OOD vs next-best



Fig. 8: Does DROID Improve Policy Performance and Robustness? We find that across all our evaluation tasks, co-training with DROID significantly improves both in distribution and OOD performance over both no co-training and co-training with the Open-X dataset. We compare success rate averaged across all tasks with standard error, and find DROID outperforms the next best method by **22%** absolute success rate in-distribution and **17%** out of distribution.

In Figure 10 we observe that using the split of the dataset with more diverse scenes yields better performance in the OOD evaluation setting. By comparing Figure 10's individual task performances with the corresponding tasks in Figure 8, we also see that the performance of co-training with the full DROID dataset matches or outperforms the performance with the subsampled dataset on all three tasks. These results suggest that the strength of DROID lies in its size and especially in

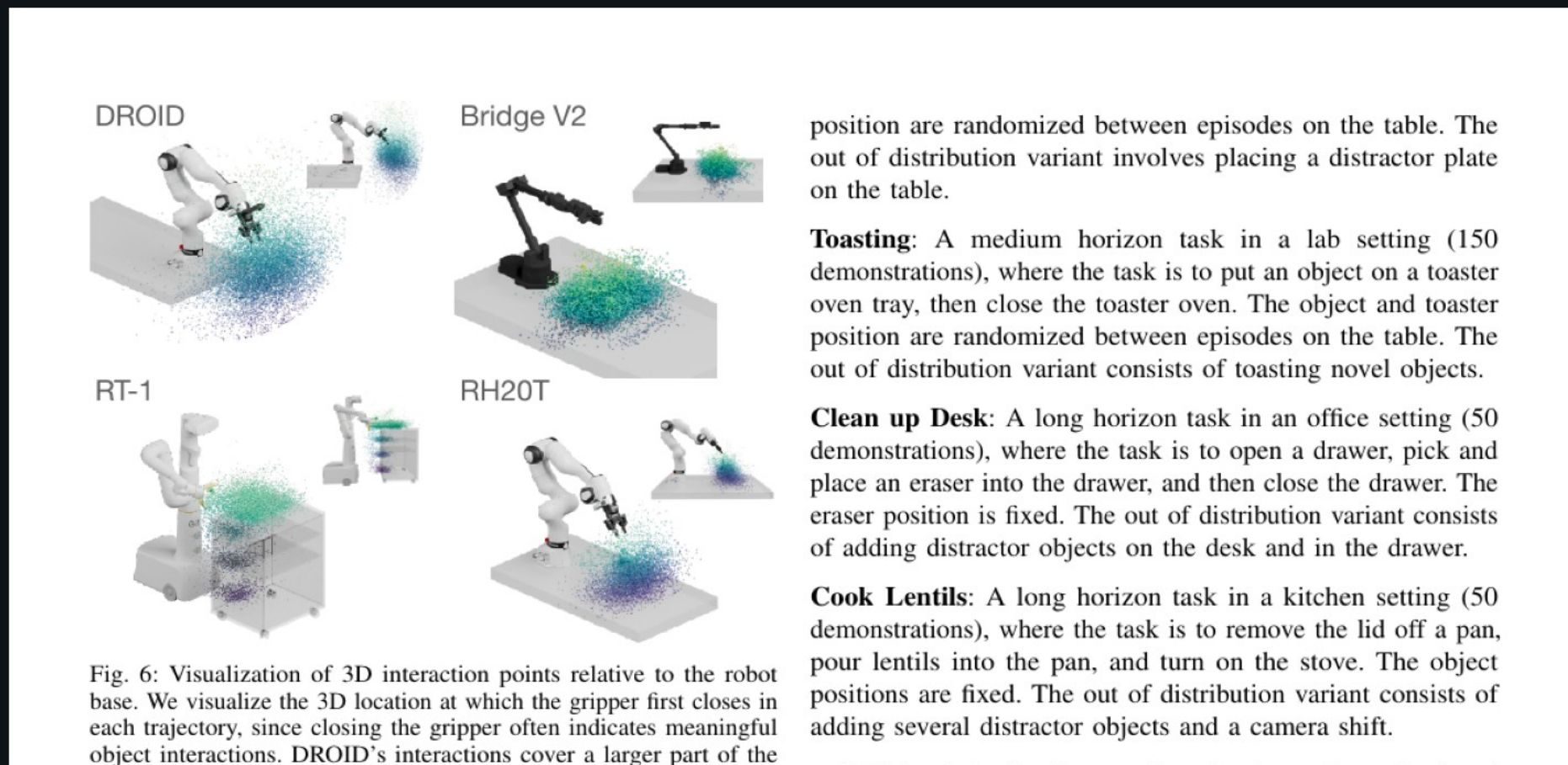
We hope that DROID can be a catalyst for research on general-purpose robot manipulation policies that are able to generalize to a broad range of tasks and scenes. In this work, we showed one example for leveraging DROID to boost policy performance, but there are many open questions about how to best make use of such diverse data: how should we combine DROID with existing large-scale robot datasets and how can we train policies that perform tasks in new scenes without any

Headlines

- ID avg: None 0.53 · OXE 0.43 · DROID 0.75
- OOD avg: None 0.17 · OXE 0.36 · DROID 0.53
- +22% ID / +17% OOD absolute
- 6 tasks × 4 locations · 10 rollouts/cell
- Co-train pool ≈ first ~40k lang-ready
- 40k ≠ headline 76k — say so
- OXE baseline = Octo soup – Language Table

Fig. 6 — interaction loci coverage

Wider 3D first-grasp workspace (not main SR)



Portable base + variable table heights → interaction-location diversity as a geometric generalization axis.

Risks / blockers

Buy data, rent modern models

Embodiment lock-in

Franka + Robotiq + ZED — not free lunch for other arms

Mix hostility

DROID can hurt underfit AR tokenizers — dip $\neq$ bad data

Wrist underused

Collected but paper policies use two exterior cams only

Pre-VLA paper stack

ResNet DP — buy *data*, rent $\pi_{0.5}$ / OpenVLA-FAST heads

Long-horizon ceiling

Cook Lentils is a demo, not a product SLA

Privacy / lang audit

Homes/offices RGB; crowd lang + GPT-4V tags need audit

BEHAVIOR steal — protocol, not river

Don't retrain 2024 ResNet DP as 2026 entry

Steal

- ✓ Scene-churn protocol (time-caps, random GUI/BDDL tasks)
- ✓ Fig. 10 matched-N diversity ablation discipline
- ✓ Diversity cards: layouts · distractors · cam jitter · instance splits

Don't

- ✗ Ship paper ResNet DP as the BEHAVIOR controller
- ✗ Drink another undifferentiated river of demos
- ✗ Ignore action-rep $\times$ fit (OpenVLA drop lesson)

Exam bite — pin this paragraph

Nov 6 midterm

“DROID shows same-robot, many-real-scenes teleop + protocolized diversity beats co-training on broader but shallower OXE pools for Franka robustness (+22%/+17%); later foundation models both depend on and sometimes drop DROID depending on capacity and action representation.”

Supporting tattoos: 564/52 · Fig.10 before Fig.8 · 2-16-8 @15Hz · $\sim 40k \neq 76k$ · π_0 9.1%

Also ready: Cook Lentils stage story · OpenVLA drop $\neq$ bad data · BEHAVIOR steals protocol not DP weights

Numbers tattoo — pin these

Nov 6 midterm ready

76k

successful traj

350 h

teleop hours

564

scenes

52

buildings

86

task verbs

50

collectors

13 / 18

labs / robots

+22% ID

Fig. 8 lift

+17% OOD

Fig. 8 lift

7362

Fig. 10 matched N

2 / 16 / 8

obs/pred/act @15Hz

~40k $\neq$ 76k

co-train pool

Stakeholder — close

Takeaways + exam bite

1. Same Franka, many real scenes (564/52) beats shallow multi-embodiment pools for Franka OOD.
2. Cite Fig. 10 before Fig. 8 for causality; tattoo +22% ID / +17% OOD and $\sim 40k \neq 76k$.
3. Motion = chunked reactive DP 2-16-8 @15Hz — Cook Lentils is stage stress, not a planner.
4. Steal protocol into BEHAVIOR on $\pi_{0.5}/N1.7$; don't ship 2024 ResNet DP as 2026 entry.

EXAM BITE

Same-robot many-scenes + protocol > broader shallow OXE for Franka (+22/+17); later models depend on & sometimes drop DROID.